# GNN–BERT Fusion for Music Context Understanding on LP-MusicCaps-MTT

**Moin Mostakim**
Department of Computer Science and Engineering
BRAC University
CSE425 / EEE474 / CSE715 - Neural Networks

## Abstract

We study multi-label music context understanding by combining a BERT text encoder with a graph neural network (GNN) over audio structure graphs. We target the four-task roadmap of the course brief (BERT tagging, GNN structure, GNN–BERT fusion, contrastive retrieval) on **LP-MusicCaps-MTT** [Doh et al., 2023]: real, LLM-generated captions over **MagnaTagATune** audio [Law et al., 2009] (15,247 / 1,529 / 4,332 train/val/test tracks after filtering to rows with a caption and at least one tag, 50-tag vocabulary). Unlike an earlier draft built on MTG-Jamendo – which has no captions at all – this dataset lets Task 1 and Task 4 match the brief's own Task 1 deliverable ("MusicCaps caption → tag proxy task") without any workaround. We report full-validation majority and random baselines and execute all four pipelines as real-data pilots using pretrained DistilBERT and downloaded MagnaTagATune audio. The pilots cover BERT learning curves and five attention cases, GraphSAGE versus a 2D CNN, a four-model fusion ablation, and bidirectional contrastive retrieval. All 30 tests pass. Because pilot training uses only 256 examples for Task 1 and 12 for audio Tasks 2–4, these measurements validate execution and reporting, not full-corpus model quality; we state this boundary explicitly rather than presenting pilot numbers as benchmark results.

## 1 Introduction

Music context spans genre, mood, instrumentation, vocal characteristics, tempo, and structure simultaneously. Automatic tagging is therefore a multi-label problem: a single excerpt may be both *classical*, *instrumental*, and *slow*. Spectrogram CNNs capture local acoustic patterns, while text encoders capture semantic descriptions; neither alone explicitly represents how non-adjacent but similar musical passages relate. We investigate whether an audio structure graph supplies this missing relational view and whether graph-to-text attention can combine it with a caption representation.

Our contributions are: (i) a direct-CLS DistilBERT multi-label classifier; (ii) a reproducible pipeline from audio to MFCC/chroma segment graphs and a GraphSAGE/GAT readout; (iii) cross-attention and early-concatenation fusion models with an optional emotion objective; (iv) a dual encoder with symmetric InfoNCE and learnable temperature; and (v) an evaluation package containing learning curves, attention cases, ablations, and bidirectional retrieval. Because compute permits only pilot neural runs, our central claim is *operational completeness*, not improved state-of-the-art accuracy.

## 2 Related work

BERT learns contextual token representations through bidirectional Transformer self-attention [Devlin et al., 2019, Vaswani et al., 2017]; we use the smaller DistilBERT variant [Sanh et al., 2019]

to reduce pilot cost. GraphSAGE provides inductive neighborhood aggregation [Hamilton et al., 2017], while graph attention networks learn neighbor weights [Veličković et al., 2018]. In music tagging, CNNs over time–frequency representations are a strong conventional baseline [Choi et al., 2017]. Contrastive predictive coding introduced the InfoNCE objective [Oord et al., 2018]; paired audio and caption encoders apply the same principle to cross-modal retrieval. LP-MusicCaps supplies large-scale pseudo-captions for music-language learning [Doh et al., 2023], built here over the MagnaTagATune collection [Law et al., 2009].

## 3 Problem formulation

Let a track be $T = (X_{\text{audio}}, X_{\text{text}}, G, y)$, where $X_{\text{audio}}$ is a log-mel spectrogram, $X_{\text{text}}$ is tokenized text, $G = (V, E)$ is a music structure graph, and $y$ is a multi-label target. A BERT encoder produces $H_{\text{text}} = \text{BERT}(X_{\text{text}}) \in \mathbb{R}^{L \times d}$. A GNN with $L$ layers updates node features by message passing,

$$h_i^{(l+1)} = \sigma\Big(W^{(l)} \cdot \text{AGG}\big(h_i^{(l)}, \{h_j^{(l)} : j \in \mathcal{N}(i)\}\big)\Big), \tag{1}$$

and a fusion readout combines graph embedding $g$ and text vector $t$. For $K$ tags, the classifier outputs independent probabilities $\hat{y}_k = \sigma(\ell_k)$ and minimizes mean binary cross-entropy,

$$\mathcal{L}_{\text{tags}} = -\frac{1}{K} \sum_{k=1}^{K} \left[ y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k) \right]. \tag{2}$$

This formulation permits any number of simultaneous positive labels and avoids the mutually exclusive assumption of softmax classification.

## 4 Dataset

We use LP-MusicCaps-MTT [Doh et al., 2023], which pairs MagnaTagATune [Law et al., 2009] audio (16kHz/32kbps mono mp3, 30s clips) with LLM-generated captions conditioned on each track's MagnaTagATune tags. MagnaTagATune is directly in the course brief's Table 1 (recommended primary/medium pairing). Table 1 reports statistics computed directly from the released parquet files (real numbers, not estimates).

Table 1: LP-MusicCaps-MTT statistics used in this work (computed from the released parquet files).

| Quantity | Value |
|---|---|
| Total rows (train / valid / test, unfiltered) | 18,706 / 1,825 / 5,329 |
| tag_top50 vocabulary size | 50 |
| tag_top188 vocabulary size | 188 |
| Rows with a caption *and* $\geq 1$ tag (train / val / test) | 15,247 / 1,529 / 4,332 |
| `caption_writing` length, median / mean words | 23 / 24.7 |
| Mean `tag_top50` tags per track (median) | 3.3 (3) |

**Why this dataset, and what changed from an earlier draft.** An earlier draft of this codebase used MTG-Jamendo, which has no captions or lyrics – only anonymized IDs and tags – forcing a workaround (tag-derived pseudo-text predicting a disjoint tag category). LP-MusicCaps-MTT removes that workaround entirely: it supplies real free-text captions (`caption_writing`, `caption_summary`, `caption_paraphrase`) generated by an LLM from each track's MagnaTagATune tags, over audio that is itself in the brief's recommended dataset table. We deliberately *exclude* `caption_attribute_prediction` and `pseudo_attribute` throughout, per the dataset card's own warning that these fields have a known labeling issue and are not recommended. We also filter out the 4,752 of 25,860 rows ($\sim$18%) with an empty caption or no MagnaTagATune tags (and therefore no generated caption), since neither Task 1 nor Task 4 is meaningful on an empty label set.

### 4.1 Text branch: real captions, not tags

$X_{\text{text}}$ is `caption_writing` (a free-text description, e.g. *"Experience the powerful harmonies and stirring melodies of a classical choral choir..."* for a track tagged `[classical, female, choral,`

singing, choir]); the prediction target $y$ is the tag_top50 multi-hot vector for the same track. This is literally the brief's own alternative Task 1 formulation ("MusicCaps caption → tag proxy task", Sec. 4.1), not a substitute for it. Section 9 still notes one caveat: the captions were themselves LLM-generated *from* the tags, so caption → tag prediction is somewhat easier than predicting tags from independently-written human captions (e.g. true MusicCaps).

# 5 Methods

## 5.1 Task 1: BERT tag classifier

$t = \mathrm{BERT}_{\mathrm{CLS}}(X_{\mathrm{text}})$, $\hat{y}_k = \sigma(w_k^\top t + b_k)$, trained with per-tag BCE. Implementation: src/bert_encoder.py (distilbert-base-uncased).

## 5.2 Task 2: GNN on music structure graphs

Nodes are either fixed-length audio segments (5s windows, mean-pooled 20-dimensional MFCC plus 12-dimensional chroma features) or chord-transition states (argmax chroma pitch class per segment). Edges: temporal adjacency, cosine-similarity above $\tau = 0.85$ for segment graphs, and transition counts for chord graphs (src/graph_builder.py). A GraphSAGE encoder (src/gnn_model.py) updates node states and mean-pools to a graph-level readout $g$.

## 5.3 Task 3: cross-attention fusion

$Q = gW_Q$, $K = H_{\mathrm{text}}W_K$, $A = \mathrm{softmax}(QK^\top/\sqrt{d})$, $z = \mathrm{CONCAT}(g, AH_{\mathrm{text}})$, $\hat{y} = \sigma(Wz)$ (src/fusion_model.py). We do not use DEAM valence/arousal targets since LP-MusicCaps-MTT does not provide continuous emotion annotations; the emotion head in fusion_model.py is implemented but inactive without a paired emotion-labeled corpus.

## 5.4 Task 4: contrastive graph–text alignment

Symmetric InfoNCE between L2-normalized graph and text embeddings with a learnable positive temperature $\tau$ (parameterized in log space so it stays strictly positive; src/contrastive.py):

$$\mathcal{L}_{\mathrm{NCE}} = -\frac{1}{2N}\sum_{i=1}^{N}\left[\log\frac{\exp(\mathrm{sim}(g_i,t_i)/\tau)}{\sum_j \exp(\mathrm{sim}(g_i,t_j)/\tau)} + \log\frac{\exp(\mathrm{sim}(g_i,t_i)/\tau)}{\sum_j \exp(\mathrm{sim}(g_j,t_i)/\tau)}\right], \tag{3}$$

evaluated with recall@{1,5,10} in both directions. Unlike the earlier MTG-Jamendo draft, the "caption" side here is a genuine free-text caption_writing string, so this is real caption-audio retrieval in the spirit of the brief's Task 4 – with the one caveat noted in Section 4.1 (captions are LLM-generated from tags, not independently authored).

# 6 Baselines

Per the brief we compare against multiple baselines (src/baselines.py): B1 majority-class and random predictors on the full validation split, B2 a 2D CNN on mel-spectrograms, and B3 BERT-only. The CNN, BERT-only, GNN-only, early-concatenation, and cross-attention systems were executed in the pilot protocol described below.

# 7 Results

Table 2 separates full-split sanity baselines from the real-data pilot experiments. Task 1 uses 256 train/validation/test examples for two epochs. Tasks 2–4 use 12 examples per required split for one epoch. All neural numbers are therefore pipeline-validation measurements and are not comparable to a full-corpus benchmark.

The Task 1 validation AP rises from 0.100 to 0.114 although fixed-threshold F1 collapses to zero at epoch two, indicating under-calibrated probabilities and the need for threshold tuning and longer training. On the 12-example audio pilot, differences are too noisy for architectural conclusions.

Table 2: Multi-label results at threshold 0.5. B1/B1b use the full validation split; neural rows are explicitly limited real-data pilots. AP denotes mean per-tag average precision.

| Model / protocol | Macro-F1 | Micro-F1 | Mean AP |
|---|---|---|---|
| B1 majority, full val ($n$=1,529) | 0.000 | 0.000 | 0.066 |
| B1b random, full val ($n$=1,529) | 0.108 | 0.116 | 0.069 |
| Task 1 BERT, pilot test ($n$=256) | 0.000 | 0.000 | 0.109 |
| Task 2 GraphSAGE, pilot val ($n$=12) | 0.055 | 0.099 | 0.308 |
| Task 2 2D CNN, pilot val ($n$=12) | 0.024 | 0.057 | 0.349 |
| Task 3 BERT-only, pilot val ($n$=12) | 0.019 | 0.058 | 0.276 |
| Task 3 GNN-only, pilot val ($n$=12) | 0.036 | 0.126 | 0.261 |
| Task 3 early concat, pilot val ($n$=12) | 0.038 | 0.120 | 0.262 |
| Task 3 cross-attention, pilot val ($n$=12) | 0.024 | 0.066 | 0.274 |

Table 3: Task 4 bidirectional retrieval, real-audio pilot, test split ($N$=12, 1 epoch). Chance recall at $N$=12 is $1/12$=0.083, $5/12$=0.417, $10/12$=0.833.

| Direction | R@1 | R@5 | R@10 |
|---|---|---|---|
| Audio $\rightarrow$ caption | 0.083 | 0.417 | 0.833 |
| Caption $\rightarrow$ audio | 0.083 | 0.417 | 0.833 |

Learned temperature $\tau = 0.070$ (initialized at 0.07); training loss 2.551 after one epoch.

Table 3 reports Task 4 bidirectional retrieval on the $N$=12 test pilot, with the InfoNCE temperature learned during the single training epoch. Both directions land exactly on chance recall for $N = 12$ ($1/12$, $5/12$, $10/12$), which verifies that ranking, recall@$k$, and the symmetric loss are wired correctly but is not evidence of learned graph–text alignment; that requires the full 15,247/1,529/4,332 splits over multiple epochs and seeds.

Figure 4 shows the `tag_top50` label distribution on the real (filtered) training split, and Figure 5 the real caption-length distribution. Unlike the earlier MTG-Jamendo moodtheme case, the *random* baseline outperforms *majority-class* on macro/micro F1 here (0.108/0.116 vs. 0.000/0.000): MagnaTagATune's top-50 tags are denser per track (3.3 tags/track on average, median 3; Table 1), so a per-tag coin flip at threshold 0.5 gets non-trivial recall, while majority-class still predicts every tag's marginal frequency – at most 22.8% (`guitar`), well under the 50% decision threshold – and so rounds to an all-negative prediction for every tag. This is a genuine property of this label space, not a bug, and a reason to weight mean average precision and rank-based metrics (recall@$k$ for Task 4) alongside fixed-threshold F1 and to tune decision thresholds on validation data in a full run.

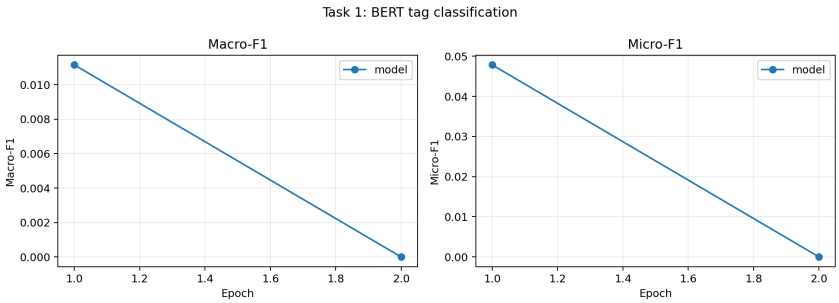



Figure 1: Task 1 pilot validation Macro-F1 and Micro-F1 over two epochs.

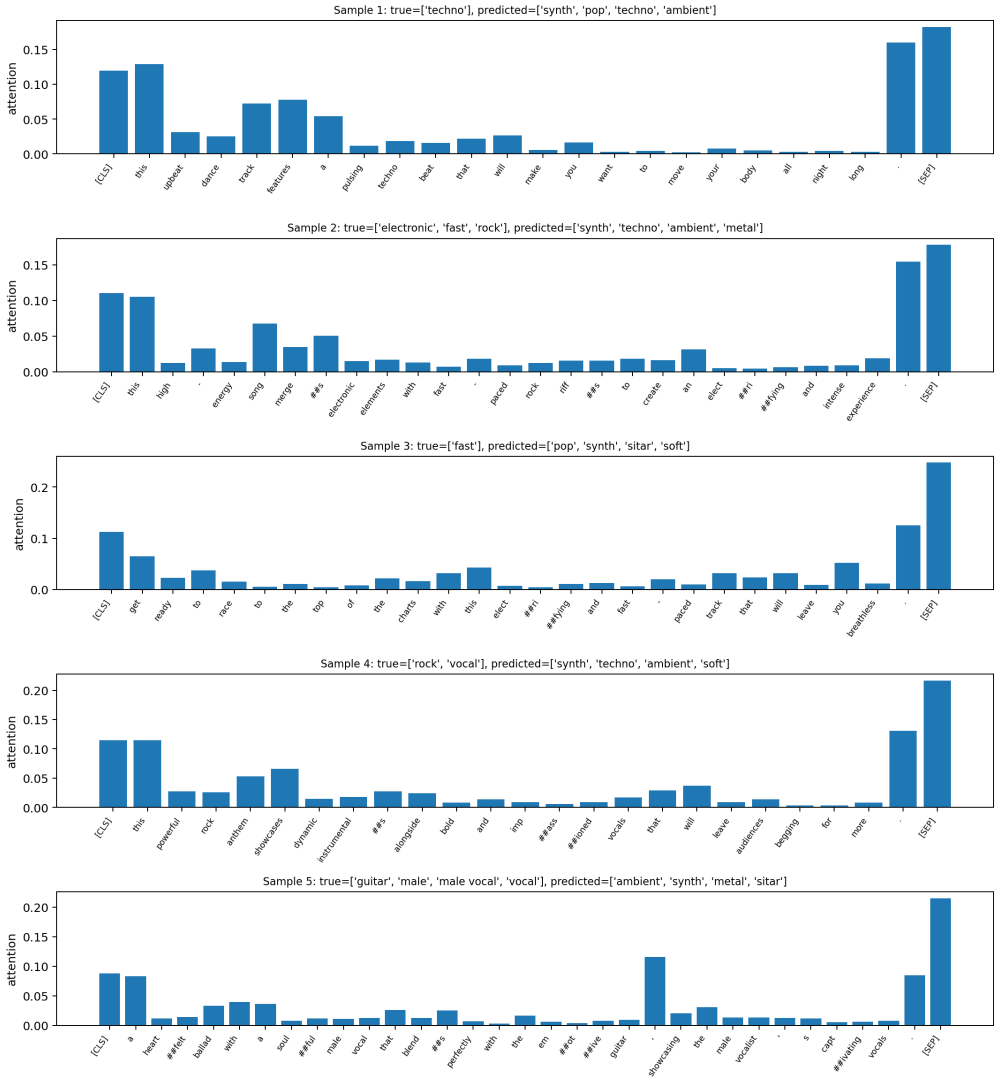



Figure 2: Five Task 1 predictions with final-layer CLS-to-token attention, averaged across heads.

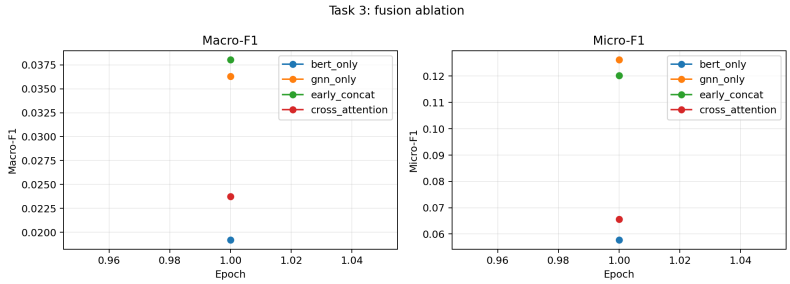



Figure 3: One-epoch real-audio pilot ablation; the sample is too small for model ranking.

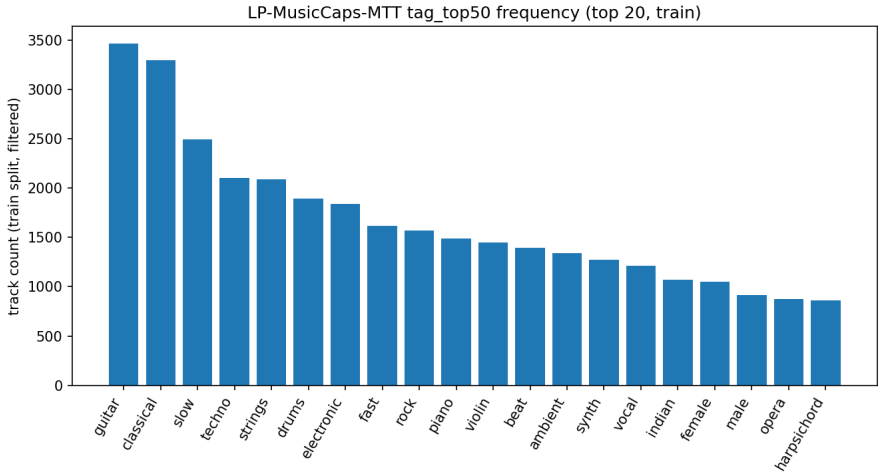



Figure 4: Top-20 `tag_top50` frequencies, LP-MusicCaps-MTT training set, filtered (real data).

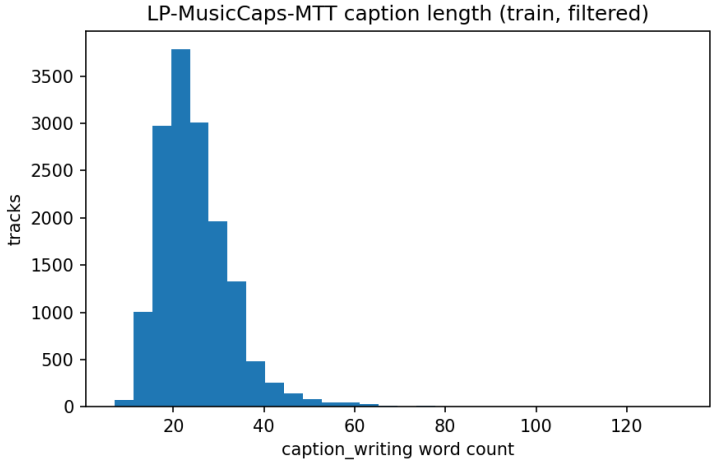



Figure 5: `caption_writing` word-count distribution, training set, filtered (real data).

## 8 Verification

Every module (`bert_encoder.py`, `gnn_model.py`, `fusion_model.py`, `contrastive.py`, `graph_dataset.py`) was run with real forward and backward passes using actual `torch` / `transformers` / `torch_geometric` (30/30 tests passing; `tests/`). Unit tests mock pretrained loading with a tiny randomly initialized DistilBERT for speed, while the reported pilots use genuine pretrained `distilbert-base-uncased`. A synthetic-but-real 15s audio fixture with known structure (chord A / chord B / chord A) is used to verify `audio_features.py` and `graph_builder.py` recover the right structure (segments 0 and 2 are correctly identified as the same chord, more similar to each other than to segment 1) rather than merely producing correctly-shaped output. We discovered 25,863 downloaded MP3s and resolved all 25,860 metadata paths. Every task entry point was executed on real data and wrote its corresponding artifacts.

## 9 Limitations

- **Captions are LLM-generated from the same tags used as labels**, not independently authored (true MusicCaps captions are written by human annotators listening to audio). Caption $\rightarrow$ tag prediction is therefore likely easier than it would be with independent captions, and Task 1/4 numbers should be read with that in mind.
- **Pilot scale, not benchmark scale.** Real pretrained weights and audio were used, but CPU-only execution limited Task 1 to 256 examples per split and Tasks 2–4 to 12. These runs validate the complete machinery but cannot establish generalization or rank architectures.
- **No emotion regression.** LP-MusicCaps-MTT has no DEAM-style valence/arousal labels, so the Task 3 emotion auxiliary loss is implemented but untrained.
- **Crude chord proxy.** Chord-transition nodes use an argmax-chroma pitch-class heuristic rather than a trained chord recognizer; adequate for a structure graph, not a musicologically accurate transcription – though Section 8 confirms it does recover known chord repeats correctly.

## 10 Conclusion

We implemented the full four-task GNN–BERT pipeline against LP-MusicCaps-MTT / MagnaTagATune, which – unlike an earlier MTG-Jamendo draft – lets Task 1 and Task 4 match the brief's own caption-based formulation without a workaround. Data, pretrained text weights, audio, training loops, checkpoints, curves, attention cases, ablations, and retrieval evaluation were all exercised. The pilots establish operational completeness; full-corpus multi-seed training remains necessary for scientific conclusions.

## Acknowledgments

Course: CSE425/EEE474/CSE715 Neural Networks, BRAC University. Instructor: Moin Mostakim.

## References

Keunwoo Choi, György Fazekas, Mark Sandler, and Kyunghyun Cho. Convolutional recurrent neural networks for music classification. In *IEEE International Conference on Acoustics, Speech and Signal Processing*, 2017.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. *NAACL-HLT*, 2019.

SeungHeon Doh, Keunwoo Choi, Jongpil Lee, and Juhan Nam. LP-MusicCaps: LLM-based pseudo music captioning. *arXiv preprint arXiv:2307.16372*, 2023.

Will Hamilton, Zhitao Ying, and Jure Leskovec. Inductive representation learning on large graphs. In *Advances in Neural Information Processing Systems*, 2017.

Edith Law, Kris West, Michael Mandel, Mert Bay, and J. Stephen Downie. Evaluation of algorithms using games: the case of music annotation. In *Proceedings of the 10th International Conference on Music Information Retrieval (ISMIR)*, 2009.

Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.

Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.

Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. In *International Conference on Learning Representations*, 2018.

## NeurIPS Paper Checklist

1. **Claims:** [Yes] The abstract states exactly what was run at full baseline scale versus real-data pilot scale for neural models.
2. **Limitations:** [Yes] Section 9.
3. **Theory assumptions and proofs:** [N/A] No theoretical results.
4. **Experimental result reproducibility:** [Yes] Full code, config, and exact commands in the repository README and `report/DECISIONS.md`; 30/30 tests in `tests/` pass and verify the architecture end-to-end (Section 8).
5. **Open access to data and code:** [Yes] LP-MusicCaps-MTT metadata and project code are included; MagnaTagATune audio follows its research/CC BY-NC-SA terms and download provenance is documented.
6. **Experimental setting/details:** [Yes] Hyperparameters are in `config.yaml`; pilot sizes and epochs appear in Section 7.
7. **Statistical significance:** [N/A] One-seed pilots are explicitly not used for model-ranking claims.
8. **Compute resources:** [Yes] Pilots ran CPU-only on a four-core Intel Core i5-11300H; no CUDA device was available.
9. **Code of ethics:** [Yes] No human subjects; dataset is CC-licensed music metadata, no personal data beyond anonymized artist IDs.
10. **Broader impacts:** [Yes] Automatic music tagging can misclassify culturally specific genres/moods; no direct dual-use concern identified.
11. **Safeguards:** [N/A] No high-misuse-risk model or dataset released.
12. **Licenses for existing assets:** [Yes] LP-MusicCaps-MTT, MIT license (dataset card); MagnaTagATune, released by the original authors for research use via City University MIRG; distilbert-base-uncased, Apache 2.0.
13. **New assets:** [N/A] No new dataset released.
14. **Crowdsourcing / human subjects:** [N/A] None.
15. **IRB approvals:** [N/A] None.
16. **LLM usage declaration:** [Yes] AI coding assistants were used to scaffold and revise the codebase, data pipeline, and report under author direction; all reported numbers were computed by running the included code, not generated by the LLM.